

Feature Comparison of Some Popular Boards

Board name	Processor	Power	RAM	EEPROM/ Flash	Storage	Peripherals	Extras
GR-Sakura	Rx63N @ 100 MHz	NA	128 kB	1 MB	MicroSD card	Ethernet, USB 2.0, OTG supported, ADC/DAC supported by wiring, headers for XBEE	JTAG, E1 ready, Power Jack, USB power
Arduino Mega	ATMega1280 @ 16 MHz	0.2W	8 kB	4 kB	128 kB	54 digital I/O pins, 16 analogue inputs, 4 UARTS	USB, power jack
Arduino Duemilanove	ATMega328 @ 16 MHz	0.2W	2 kB	1 kB	32 kB	14 digital I/O pins (six with PWM), eight analogue input pins	Powered via USB, 7-12 V
bed NXP LPC1768	ARM Cortex-M3 @ 96 MHz	NA	32 kB	512 kB	NA	40-pin DIP, 2xSPI, 2xI2C, 3xUART, CAN, 6xPWM, 6xADC, GPIO	Breadboard mountable, Ethernet, USB drag 'n' drop Flash programmer, USB host/device
TI MSP430 LaunchPad	MSP430 @ 25 MHz	0.28 W	512 bytes	None	16 kB	10 GPIO	EEPROM emulation, 0.5m USB cable
STM32 Discovery	Cortex-M3 @ 72 MHz	NA	8 kB	4 kB	128 kB	Extension header for all QFP64 I/Os, STLink on-board	Two pushbuttons user and reset, two user LEDs
chipKIT Uno32	PIC32 @ 80 MHz	NA	16 kB	None	128 kB	16-/32-bit timers, PWM, 16-channel 10-bit ADC, comparators, I2C, SPI, UART	Same form-factor as Arduino Uno, compatible with most Arduino shields
BeagleBoard	ARM Cortex-A8 @ 600 MHz	2 W	256 MB	—	256 MB	USB on the go, RS-232, JTAG, 3.5mm jacks, MMC slot with SDIO, DVI-D, S-video	Alternative boot source button, boot from NAND, MMC, USB or serial
Atmel STK500	AVR	NA	NA	NA	2 MB	RS-232	Parallel and serial high-voltage mode, compatible with all Atmel chips
Raspberry Pi	ARM11 @ 700MHz	2.5W-3.5W	256 MB	—	MicroSD	3.5mm jack, HDMI, USB 2.0, Ethernet RJ45	GPU handGPU handles 1080p video

efficient Rx63N MCU that is clocked at 100 MHz (165 DMIPS). It comes with 128kB RAM and a full megabyte of Flash. The board features Arduino-compatible headers, which makes it compatible with many forms of Arduino shields.

Renesas has worked to make GR-Sakura an easy-to-set-up board for the user to easily design and create original ideas for electronic gadgets. Plug in some sensors, develop code to manipulate the input, connect some output devices and you are done. In case you are not really interested in sitting in front of a PC punching in code, you can utilise another feature of this board that allows you to easily program the board through a smartphone.

"The Sakura board works not just with Windows but also with an Apple Mac computer"

There is one important difference between Arduino boards and Sakura—Sakura runs on 3.3 V, while Arduino uses a 9V power supply. Keep this in mind while using the shield in order to ensure that you do not fry the device.

Getting started

Setting up GR-Sakura is quite easy. You can begin by plugging in the mini-USB cable to your PC. This cable functions as both the power supply and the data transfer cable for the board. The switch will flicker and the blue LEDs start blinking to notify power connectivity.

Next, press the red reset button on the board. Once done, your PC will detect the Sakura board as a new memory drive. As you open the drive, you can find the link that takes you to the Sakura website where you will be greeted with further instructions. One interesting thing is that the Sakura board works not just with Windows but with an

Apple Mac computer too.

Once you are on the Sakura board website, you will be shown how to navigate the features and utilise the cloud Web compiler to build your program. Of course, you will have to register first before you can access this impressive tool.

Have fun!

We all love our Arduino boards and the awesome projects that we make out of them, just like we enjoyed playing with Raspberry Pi as it ran OS that previously required a full computer while providing a lot of interesting interfaces like USB, microSD and LAN at the same time. Now it seems we have a mix of both the worlds—an open board that runs a really low-power processor while allowing compatibility with both Arduino shields and a cloud-based programming environment that will help you set up in no time. ●

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